

Optimizing Study Schedule for Students using Hybrid Particle Swarm Optimization and Simulated Annealing (PSO-SA) Approach

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Abstract— Effective time management is one of the many ways students can succeed academically. However, creating a study schedule that balances each student's needs is still challenging. The difficulty of using limited resources, including time, in a way that balances competing goals and maximizes learning outcomes leads to this complexity. The importance of strategies including goal-setting, progress monitoring, and fostering a positive learning environment is highlighted by research on self-regulated learning. The benefits of scheduled study periods with breaks are demonstrated by techniques such as the Pomodoro Technique. To create an optimal study schedule that can cater to the preferences of the learner, optimization algorithms like Particle Swarm Optimization (PSO) and Simulated Annealing (SA) are expanded upon in this work. While PSO offers a robust population-based search mechanism, SA excels at refining local solutions and avoiding suboptimal traps. Understanding their complementing advantages, we suggest a hybrid PSO-SA approach to tackle the complex study schedule optimization problem. By eliminating time gaps and imbalances and balancing subject priorities, allotted study time, and breaks, the algorithm seeks to enhance study efficiency. The findings show that highly customized and productive study plans may be created by combining SA's local optimization potential with PSO's worldwide search capabilities. This method offers a useful framework for boosting academic results through improved time management by increasing efficiency and catering to personal preferences.

Keywords—Particle Swarm Optimization, Simulated Annealing, Hybrid Algorithm, Fitness Function, Study Schedule

I. INTRODUCTION

For learners to succeed academically, effective time management is essential. It becomes difficult to create a well-balanced study plan when there are many different subjects to concentrate on and several activities demanding for their attention. This difficulty arises from the requirement to distribute limited resources, including time, in a way that optimizes learning results while taking into account personal preferences and priorities.

A. Review of Related Literature

a. Study Schedule

Researches on self-regulated processes showed that the best self-regulated learning processes implement effective learning strategies, monitor and assess goal progress, establish more productive environment for learning, and more importantly maintain one's sense of self-efficacy. In order to achieve the desired outcome

of self-regulated learning process, it is necessary to emphasize the importance of self-regulated learning strategies, including planning and time allocation, in improving academic outcomes [9].

Several studies have explored methods for optimizing study schedules and time management frameworks. An example of this is the famous Pomodoro style of studying where studying is composed of 25-minute focus study then 5-minute break [8].

b. Used Baseline Algorithms

Optimization techniques have also been a focus of recent research. Kennedy and Eberhart in 1995 developed Particle Swarm Optimization (PSO) as an efficient method for population-based search, inspired by the behavior of animals and insects that live in swarms [5]. Similarly, Simulated Annealing (SA) as a robust technique for local optimization, capable of escaping local minima to find near-optimal solutions [7]. These foundational works underscore the potential of combining heuristic approaches for solving multi-criteria optimization problems like study scheduling.

B. Proposed Hybrid PSO-SA Algorithm

Creating a balanced study schedule tailored to individual preferences is a challenging optimization problem due to competing priorities and limited resources. In order to optimize study schedules, this research presents a hybrid approach that combines Simulated Annealing (SA) with Particle Swarm Optimization (PSO). By avoiding gaps and time imbalances and striking a balance between study time, breaks, and subject priorities, the suggested approach seeks to optimize study efficiency. The hybrid approach combines the advantages of both algorithms to provide customized schedules by combining SA's local optimization with PSO's population-based search.

II. METHODOLOGY

A. Particle Representations

A single particle contains the data representing the schedule of the subjects, their priorities of each subject, and the break times per subject set by the student. It also contains the information about its velocity and personal best ($Pbest_n$) schedule and fitness.

B. Fitness Function

A fitness function is used to determine which of the particles will be followed by the swarm and will be the new basis of their movements/search pattern. It shows whether the information held by that particle has the minimum fitness during the current iteration. The fitness function is defined by the equation:

$$\begin{aligned} \text{Fitness} &= -(P.T - T_p - B_p - G_p) \\ \text{where } P &= \text{priority of the subject,} \\ T &= \text{time scheduled for the subject,} \\ T_p &= \text{time imbalance penalty,} \\ B_p &= \text{break penalty,} \\ G_p &= \text{gap penalty} \end{aligned}$$

The fitness function seeks to maximize priority-weighted study time while minimizing penalties associated with imbalance, inefficient breaks, and large gaps. The negative sign ensures that the algorithm minimizes the total penalty and maximizes the overall fitness, aligning with the optimization goal.

C. Penalty Functions

Penalty functions are the key on implementing an effective fitness function for the proposed algorithm. The formula for penalizing the time, break, and gap came from the algorithms for solving constrained optimization problems, specifically using Quadratic Penalty Function.

For the time imbalance penalty, the formula given by:

$$T_p = \lambda_1 \left(\frac{\sum_{i=1}^N \sum_{j=1}^N |t_i - t_j|}{2N^2 \bar{t}} \right)^2$$

The objective is to penalize uneven allocation of the generated study time across subjects. While prioritization is important, but doing so in an unbalanced way—for example, concentrating just on one or two topics—can cause problems. The imbalance penalty encourages equity by prohibiting excessive focus on a small number of subjects during study sessions. The idea of the Gini coefficient $\left(\frac{\sum_{i=1}^N \sum_{j=1}^N |t_i - t_j|}{2N^2 \bar{t}} \right)$, which is frequently employed in economics to quantify inequality, is the source of the penalty [6]. For schedules with significant discrepancy, the penalty is increased by squaring the Gini coefficient. However, the trade-off between balance and prioritizing is controlled by the constant value λ_1 . Stricter balance is enforced by a higher λ_1 , but greater flexibility in priority is permitted by a lower λ_1 .

For the break penalty, the formula given by:

$$B_p = \lambda_2 \sum_{i=1}^N (B_i - t_i \cdot b_{\text{optimal}})^2$$

The objective is to penalize inefficient or excessive breaks across subjects. Breaks are necessary for maintaining focus and preventing burnout, but their duration should be proportional to the study time. The optimal break-to-study ratio (b_{optimal}) is based in cognitive research, which suggests that periodic breaks enhance retention and productivity (e.g., the Pomodoro technique recommends a 5-minute break for every 25 minutes of study which is a 1:5 break to study time ratio) [2]. The deviation of actual break time B_i from the expected break time ($t_i \cdot b_{\text{optimal}}$) is penalized quadratically to

discourage significant deviations. Quadratic penalties ensure that even small deviations contribute noticeably to the penalty. The constant λ_2 is similar to λ_1 which controls the penalty.

For the gap penalty, the formula given by:

$$G_p = \lambda_3 \sum_{i=1}^{N-1} (t_i - t_{i+1})^2$$

The objective is to minimize large gaps between study sessions. Large idle gaps between study sessions can lead to inefficiencies and disrupt the study flow. A compact schedule, where consecutive sessions are well-aligned, maximizes productivity. By penalizing the squared differences between consecutive study times ($t_i - t_{i+1}$) the gap penalty encourages smoother transitions. The quadratic term emphasizes larger gaps more strongly than smaller ones. The constant λ_3 is similar to other constants which controls the penalty.

D. Proposed PSO-SA Hybrid Algorithm

Algorithm 1 Algorithm for Generating Study Schedule

function PSO-SA(schedule, FITNESS)

inputs: N, number of subjects

P, subject priorities

B, break times

S, study time per day

FITNESS, fitness function

For each particle p

For each subject in schedule

time_{subject} ← random number from 0 to S / N

velocity_{subject} ← 0

End for

Normalize the schedule

p ← schedule, velocities

swarm ← p

End for

Iteration i=0

Do

For each particle p

fitness ← FITNESS(schedule_p)

If fitness < Pbest_p

Pbest_p ← schedule, fitness

End if

End for

Gbest_i ← Particle having the best fitness value

Do

Generate a neighbor particle with a random change in

time from 0 to S on a random subject

Normalize the schedule of the neighbor

If fitness_{neighbor} < fitness_{Gbest_i}

or random() < metropolis function

Gbest_i ← neighbor

End if

T = cooling_rate x T

While T is greater than T_{min}

For each particle p

For each subject in schedule

velocity_{subject} ← calculate the new velocity

time_{subject} ← time_{subject} + velocity_{subject}

Normalize the schedule

End for

End for

$i=i+1$

While maximum iteration is not attained

returns Gbest _{i}

III. RESULTS AND DISCUSSION

The proposed hybrid PSO-SA algorithm was tested using data preferences from different students during their exam week. A total of seven (7) students from different programs tested the algorithm to generate their schedules:

TABLE I. STUDENT PREFERENCES

Student 1	
N	5
P	Subject 1: 3, Subject 2: 1, Subject 3: 4, Subject 4: 5, Subject 5: 2
B	All 10 minutes
S	10 hours
Student 2	
N	6
P	Subject 1: 4, Subject 2: 2, Subject 3: 1, Subject 4: 3, Subject 5: 5, Subject 6: 6
B	All 15 minutes
S	8 hours
Student 3	
N	4
P	Subject 1: 4, Subject 2: 2, Subject 3: 3, Subject 4: 1
B	All 20 minutes
S	6 hours
Student 4	
N	7
P	Subject 1: 6, Subject 2: 2, Subject 3: 4, Subject 4: 3, Subject 5: 1, Subject 6: 5, Subject 7: 7
B	All 12 minutes
S	9 hours
Student 5	
N	3
P	Subject 1: 2, Subject 2: 3, Subject 3: 1
B	All 10 minutes
S	5 hours
Student 6	
N	6
P	Subject 1: 5, Subject 2: 4, Subject 3: 6, Subject 4: 3, Subject 5: 2, Subject 6: 1
B	All 8 minutes
S	9 hours
Student 7	
N	5
P	Subject 1: 2, Subject 2: 3, Subject 3: 5, Subject 4: 1, Subject 5: 4
B	All 10 minutes
S	8 hours

Table I displays the parameters, including the number of subjects, priorities, allocated study hours, and break durations for each student. The algorithm generated optimized schedules over a 7-day period, and fitness scores were calculated to evaluate the efficiency of each schedule.

A. Testing Setup

To achieve an optimal result upon the testing of the algorithm, the following constant parameters are used for the following baseline algorithm. Reference [3] as cited in [4] suggested for PSO that both social and personal learning parameters are equal to 1.494 ($c_1 = c_2 = 1.494$) and the inertia is set to 0.73 ($\omega = 0.73$) in order to guarantee convergence while encouraging exploration. The number of particles used is set to 5 and the number of iterations is set to 10. The small number of particles and number of iterations will be complemented by the application of Simulated Annealing, as it will check if the best particle is stuck in the local or global minima. For Simulated Annealing, the minimum and maximum temperature is set to 0 and 2000 respectively with a cooling rate of 95% ($t_{min} = 0; t_{max} = 2000; c = 0.05$) [1].

B. Stability Test

The schedule preference of each student was then used to generate schedules for 7 days, and here are the fitness scores graphed:

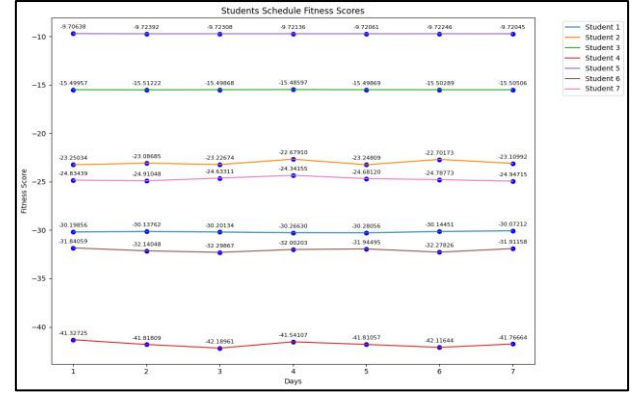


Fig. 1. Fitness Scores of the Schedule Preferences for 7 Days

As shown in the graph in Fig. 1, the fitness values of the generated schedules for 7 days are relatively similar with each other per student, showing the stability of the algorithm for multiple attempts. The simpler the preferences are, the more stable the fitness score of the generated schedule by the hybrid algorithm, and the more complex it is, the disparity is now visible but not that significant that it might alter the schedule completely.

C. Baseline Algorithms Fitness Comparison

Using the schedule preference data from the seven students, the proposed hybrid algorithm was then compared to the baseline algorithms, PSO and SA, all subjected to the same parameters:

TABLE II. FITNESS VALUES AVERAGE

Student	PSO-SA Hybrid	PSO	SA
1	-30.18586	-29.71780	-30.06017
2	-23.04325	-22.23358	-23.03500
3	-15.50044	-15.30681	-15.50115
4	-41.79567	-39.80341	-41.58100
5	-9.71975	-9.70025	-9.71407
6	-32.05951	-31.57694	-32.13787
7	-24.73366	-24.64204	-24.82206

^a. See Appendix A for specific graphical representation per student

As shown in Table II, the PSO-SA hybrid algorithm has a better average fitness than just the baseline PSO algorithm, thus making it more efficient. Compared also to the baseline SA algorithm, the hybrid algorithm has a better average fitness score on 4 out of 7 instances, and on the 3 instances where baseline SA algorithm has better average, the PSO-SA hybrid algorithm is not far from it. The simpler and more flexible the preferences are, the better for the proposed hybrid PSO-SA algorithm for generating schedules. Altering the scope of the constant parameters of the hybrid algorithms can change its performance compared to any of the baseline algorithms.

Increasing only the number of particles and max iteration to both 100 for the hybrid PSO-SA and baseline PSO will produce the following results:

TABLE III. FITNESS VALUES AVERAGE (INCREASED SCOPE FOR HYBRID PSO-SA AND BASELINE PSO ALGORITHM)

Student	PSO-SA Hybrid	PSO	SA
1	-30.24982	-30.28905	-29.87424
2	-23.26448	-23.28490	-22.98030
3	-15.51508	-15.51829	-15.49513
4	-42.31953	-42.34405	-41.41188
5	-9.72184	-9.72396	-9.70752
6	-32.38774	-32.45895	-32.03514
7	-24.94293	-24.96985	-24.69395

^a See Appendix B for specific graphical representation per student

As shown in Table III, the baseline PSO is shown to be the best out of the 3 algorithms as it capitalizes on the number of particles searching for the fittest schedule, they can find for a large number of iterations thus fixing its weakness for global searching by having multiple searching particles. The baseline SA comes in last as it heavily relies on randomness and probability in exploring globally and has a weakness in terms of local exploration. The hybrid PSO-SA algorithm is only shy behind the baseline PSO as it doesn't have its full exploration capability because of the probabilistic nature of SA.

Increasing only the maximum temperature to 40,000 for the hybrid PSO-SA and baseline SA will produce the following results:

TABLE IV. FITNESS VALUES AVERAGE (INCREASED SCOPE FOR HYBRID PSO-SA AND BASELINE SA ALGORITHM)

Student	PSO-SA Hybrid	PSO	SA
1	-30.04521	-28.74012	-29.84891
2	-23.12825	-22.62834	-23.04586
3	-15.49677	-15.41887	-15.47728
4	-41.60436	-39.91849	-41.49428
5	-9.71650	-9.68218	-9.71760
6	-32.14357	-31.73730	-32.10139
7	-24.80758	-24.41889	-24.76843

^a See Appendix C for specific graphical representation per student

As shown in Table IV, the proposed hybrid PSO-SA algorithm is shown to generate the fittest schedule among the 3 algorithms except on the schedule for Student 5 as the preferred schedule data given by Student 5 is the simplest out of all the students' preferred schedule data, thus the search space is smaller compare to other generated schedules. This also shows that baseline PSO doesn't have enough capability to get out of local minima given small number of particles, that is why in addition to baseline PSO, SA is also useful as it increases the global exploration capability of

baseline PSO which is shown in the fitness scores generated by the proposed hybrid PSO-SA algorithm.

Increasing all the constant parameters for the 3 algorithms will produce the following results:

TABLE V. FITNESS VALUES AVERAGE (INCREASED SCOPE FOR ALL ALGORITHMS)

Student	PSO-SA Hybrid	PSO	SA
1	-30.26718	-30.28905	-29.93346
2	-23.27472	-23.28490	-23.04626
3	-15.51278	-15.51829	-15.47878
4	-42.28517	-42.34404	-41.36292
5	-9.72217	-9.72396	-9.71004
6	-32.43503	-32.45895	-32.08919
7	-24.95661	-24.96985	-24.78018

^a See Appendix D for specific graphical representation per student

As shown in Table V, the baseline PSO algorithm is still the best out of the three (3) as again, it capitalizes on the large number of particles and iterations. However, comparing to the rest of the algorithms, the average fitness score isn't really that far especially with the proposed hybrid PSO-SA algorithm.

By analyzing the different scenarios of different alterations, the suggested hybrid PSO-SA algorithm is not the most optimal of the three, but it is efficient enough in terms of time and space usage to compete with the two baseline algorithms to the extent that expanding the scopes of the constant variables is the only way for them to have an advantage over the hybrid PSO-SA algorithm. Despite this, the gap between the hybrid PSO-SA compare to the baseline algorithms isn't even large enough to consider the trade-offs in computational resources.

D. Baseline Algorithms Time Comparison

Measuring the time of the three (3) algorithms during their comparison with the given alterations (1) No alteration, (2) Increased scope for hybrid PSO-SA and baseline PSO algorithm, (3) Increased scope for hybrid PSO-SA and baseline SA algorithm, and (3) Increased scope for all algorithms, resulted to these measures:

TABLE VI. AVERAGE COMPUTATION TIME

Alteration	PSO-SA Hybrid	PSO	SA
1	0.39995	0.00687	0.01771
2	4.77587	0.79443	0.01847
3	0.41197	0.00748	0.01995
4	4.80027	0.79824	0.01863

Table VI displays the computation time for the three algorithms (PSO-SA Hybrid, PSO, and SA) under different alteration conditions reveals significant differences in processing efficiency. These results show the trade-offs between computational cost and optimization performance.

The hybrid PSO-SA algorithm demonstrated a substantially greater computation time (0.39995 seconds) than PSO (0.00687 seconds) and SA (0.01771 seconds) under the baseline situation with no changes made to the parameters. Because the hybrid algorithm incorporates the methods of both PSO and SA, it naturally requires more computing, therefore this outcome is expected. Even in its default setup, the hybrid model still involves a

computational cost even though it yields higher optimization outcomes, as previously discussed.

The hybrid PSO-SA and baseline PSO algorithms' computation times increased significantly when the number of particles and maximum iterations were raised. Compared to PSO, which takes 0.79443 seconds, the hybrid PSO-SA took an average of 4.77587 seconds. Because of its dual-optimization structure, which processes a bigger search space with more iterations, the hybrid algorithm has higher analysis and calculations, which is reflected in this result. For application that requires more precision and has enough resources, baseline PSO might be a better option than the hybrid PSO-SA and SA algorithms.

In comparison to the baseline condition, the computation time for both the hybrid PSO-SA and baseline SA algorithms increased somewhat for the third alteration. SA increased to 0.01995 seconds, but the hybrid algorithm's average time climbed somewhat to 0.41197 seconds. This alteration implies that the increase in particles and iterations has a greater effect on computation time than the rise in maximum temperature. In this case, the hybrid method maintains a competitive processing efficiency with a negligible computational expense.

Both the hybrid PSO-SA and PSO algorithms' computation times increased when all three algorithms' parameters were raised; the hybrid took 4.80027 seconds, while PSO took 0.79824 seconds. At 0.01863 seconds, SA's computation time stayed comparatively constant throughout the alteration. This outcome emphasizes the hybrid and PSO algorithms' scalability issues when parameters are greatly increased. The hybrid approach is better suited for issues where computational resources are not a constraint because of its substantial computation time, which is a reflection of its complex optimization process.

A clear trade-off between computational efficiency and optimization performance is revealed by the computation time study. Because of its mixed processes, the hybrid PSO-SA algorithm constantly exhibits longer computation durations, particularly as the parameter scope is expanded. However, for issues needing high precision, the hybrid algorithm's higher optimization outcomes could offset the extra computing cost.

On the other hand, the independent PSO and SA algorithms continue to have shorter computation durations, which makes them better suited for situations with limited resources. Notably, SA shows the most consistent calculation time across all changes, indicating that increasing the parameters has less of an impact on it. Therefore, the particular needs of the problem should dictate the method selection, weighing the need for high-quality optimization against the computational resources at hand.

E. Expected Output

After inputting the schedule preference data from the students, the output will be an object containing the time generated per subject.

```
Generated Schedule = {
  0: 1.8584609629831998,
  1: 1.3595923624610868,
  2: 1.1874612872574406,
  3: 1.5944853872982723
}
```

Fig. 2. Sample Expected Output for 1 Day

With a bit of formatting, along with additional parameters for formatting (e.g. Starting Time, Time Chunking), the student will receive something similar to this:

```
Day 1
Subject 1:
  06:00 AM - 07:40 AM (100 minutes)
  (Break: 20 minutes)
  08:00 AM - 08:11 AM (11 minutes)
  (Break: 20 minutes)
Subject 2:
  08:31 AM - 09:53 AM (81 minutes)
  (Break: 20 minutes)
Subject 3:
  10:13 AM - 11:24 AM (71 minutes)
  (Break: 20 minutes)
Subject 4:
  11:44 AM - 01:20 PM (95 minutes)
  (Break: 20 minutes)
```

Fig. 3. Sample Formatted Output

The schedule shown in Fig. 3 is from the Day 1 of Student 3. Student 3 wants to start studying at 6:00 AM, 4 subjects will be studied in the span of 6 hours with 20 minutes of breaks in between based on the schedule preference data. The generated time per subject is then separated into chunks of fix maximum time based on the optimal break-to-study ratio and added the preferred breaks in between.

IV. CONCLUSION

The hybrid PSO-SA algorithm presented in this study demonstrates its capability to optimize study schedules effectively by leveraging the global search efficiency of Particle Swarm Optimization (PSO) and the local refinement ability of Simulated Annealing (SA). Through experiments with diverse student preferences, the hybrid approach consistently produced schedules with competing fitness scores compared to baseline algorithms under varying conditions and have a minimal trade-off in computational resources given specific alterations. While the hybrid PSO-SA algorithm does not consistently outperform PSO in scenarios with extensive particle exploration, it shows robust performance in balancing computational efficiency and optimization quality with the help of SA. Furthermore, it provides competitive results even in resource-constrained scenarios, showcasing its adaptability and effectiveness.

The findings of this research highlight the potential of hybrid algorithms in tackling multi-criteria optimization problems. Future work may focus on refining the algorithm further by exploring dynamic parameter adjustments and applying it to other domains requiring personalized scheduling. Also, future researchers may incorporate additional parameters to further optimize and improve the efficiency of the generated the schedules to better help the learners.

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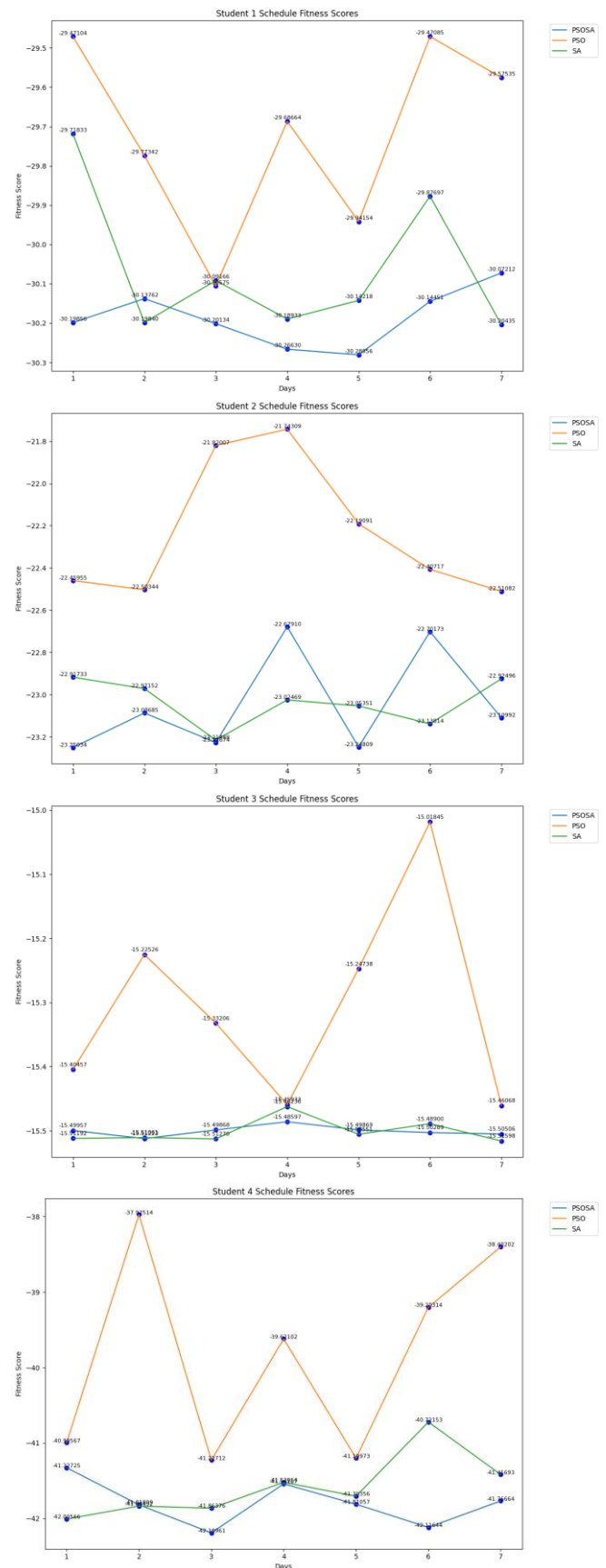
Thank you all and mabuhay!

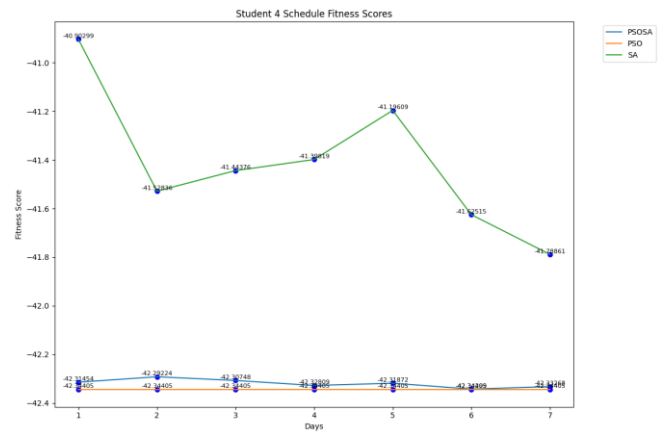
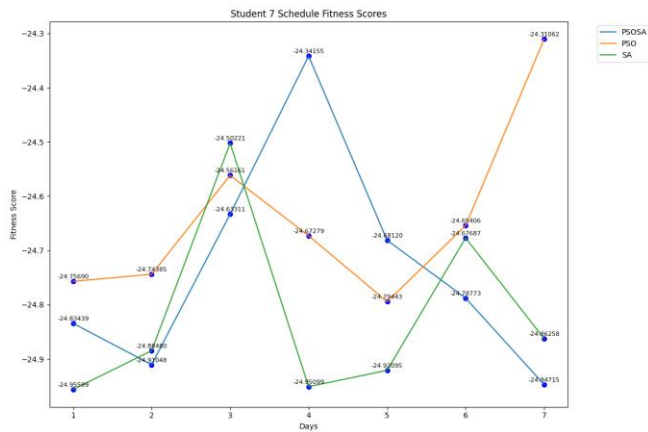
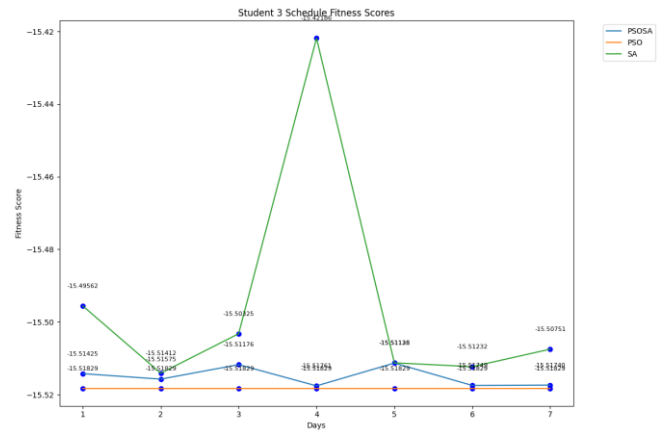
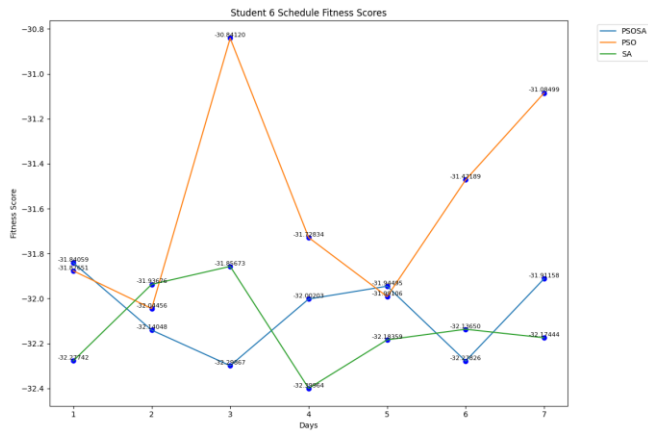
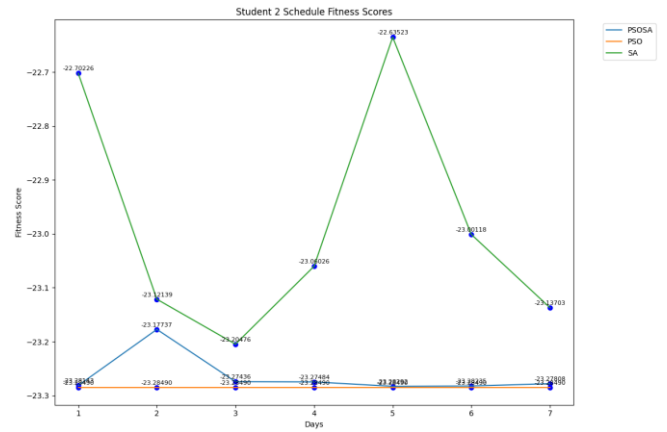
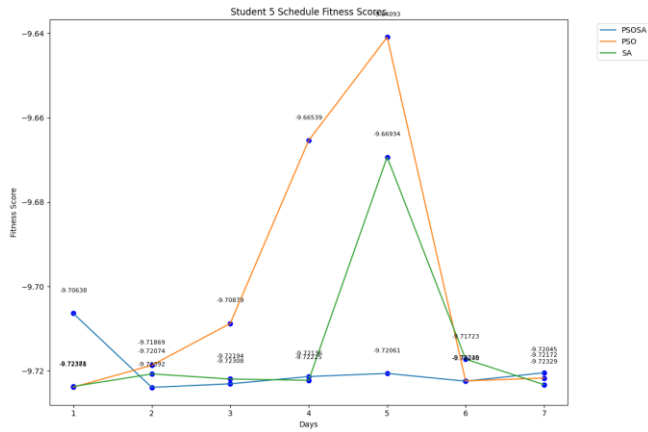
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Appendix A

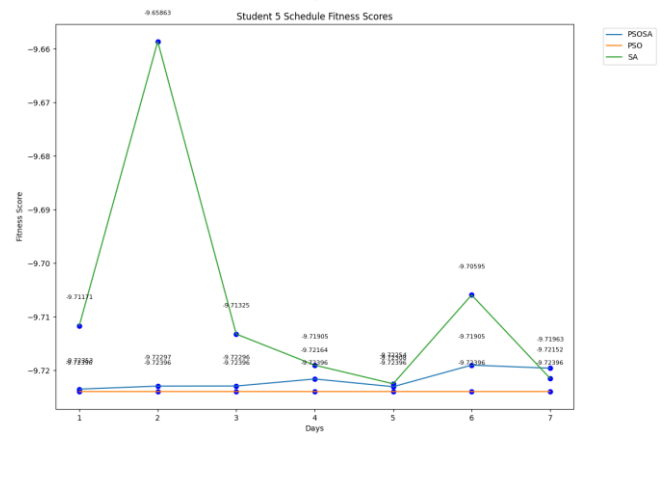
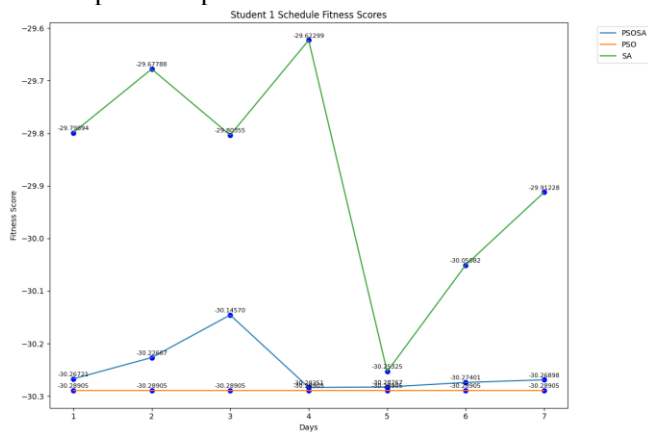
Graphs for Specific Fitness Scores for Table II

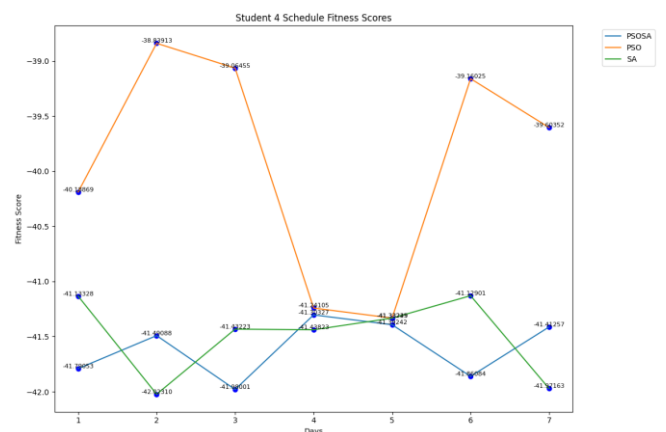
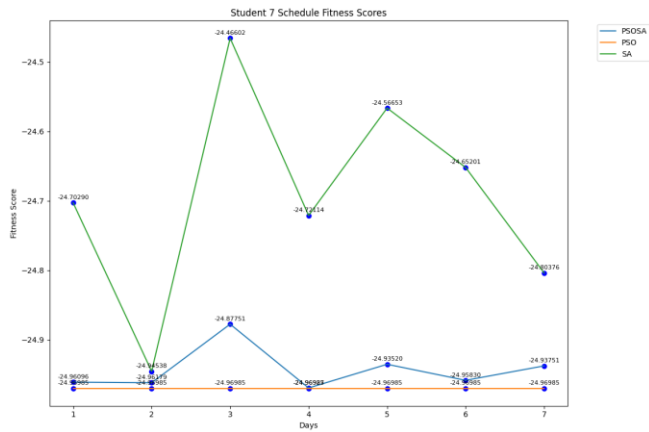
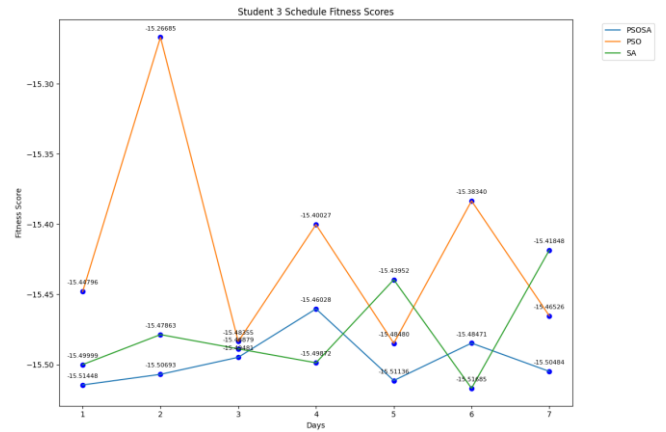
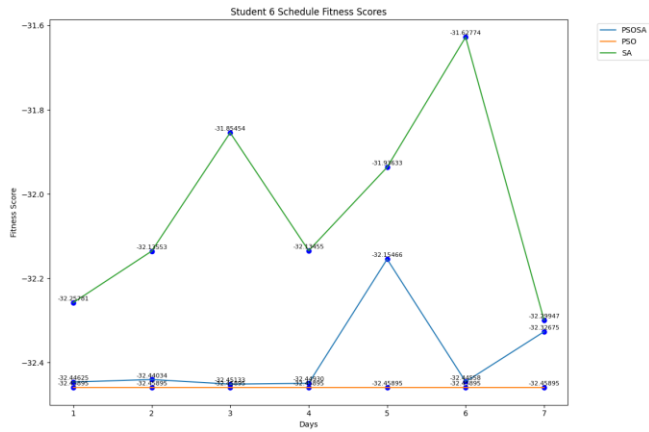




Appendix B

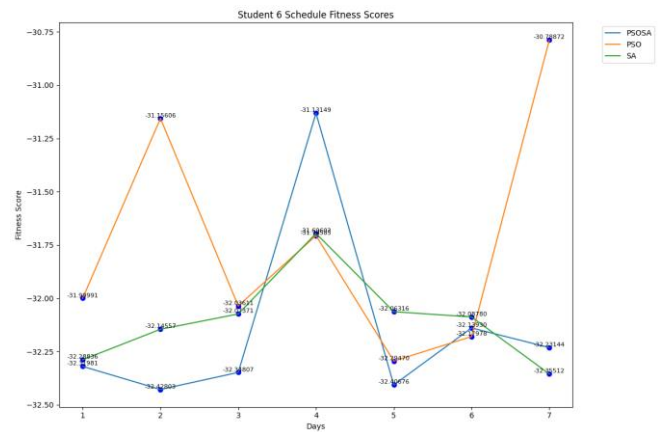
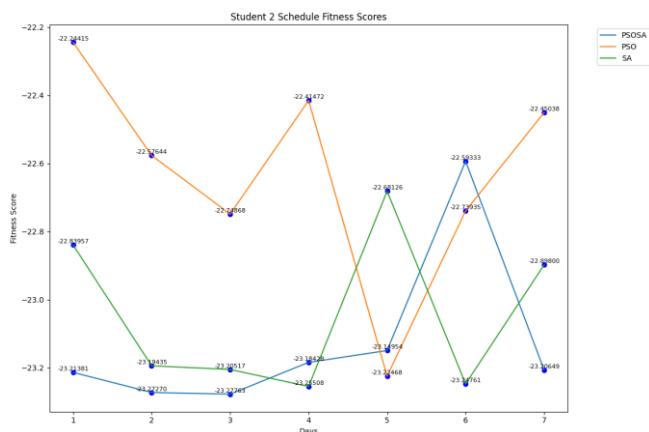
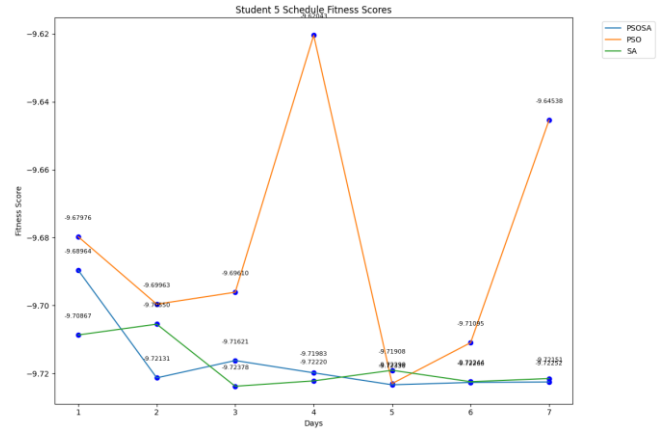
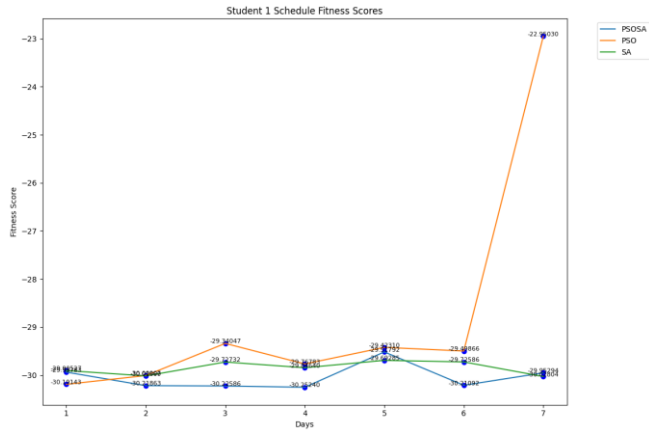
Graphs for Specific Fitness Scores for Table III

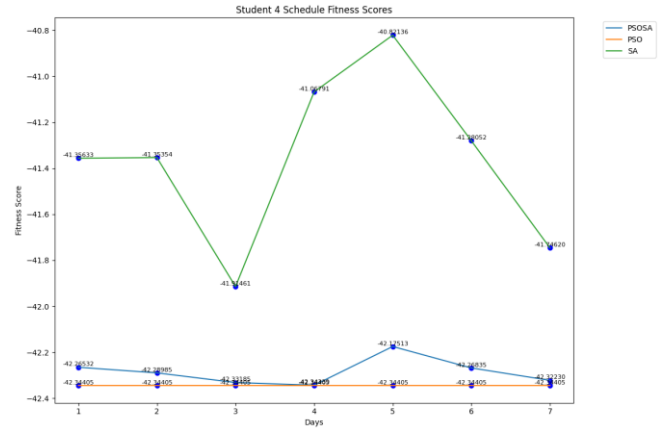
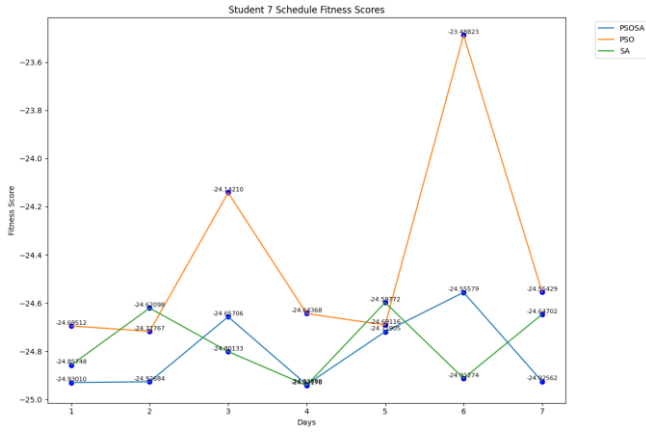




Appendix C

Graphs for Specific Fitness Scores for Table IV





Appendix D

Graphs for Specific Fitness Scores for Table V

